

time, the stronger is the synaptic connection. Essentially, it memorises the action between neurons. So it does, in fact, run a bit like a neuron, in the sense that it adapts, strengthens and weakens connections according to external stimuli.

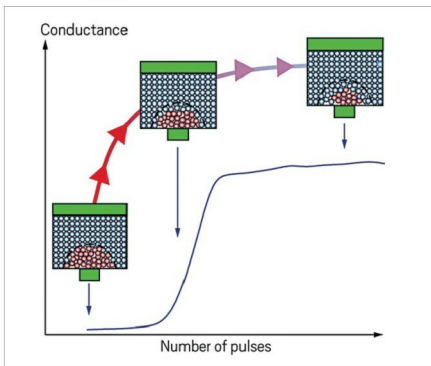
Also, opposed to traditional transistors, the creation is not restricted to the binary system of 0s and 1s and, interestingly enough, runs on non-volatile memory, which means that, even when power is interrupted, the device remembers its state. Still, it cannot form new connections like a human neuron can. By exploiting the extreme sensitivity of this material, a very small excitation allows to get a large signal, so the input energy required to drive this switching is potentially very small. That could translate into a large boost for energy efficiency. It does have a significant advantage over the human brain—these transistors can run at high temperatures exceeding 160 degree Celsius. This kind of heat typically boils the human brain.

Technological developments

The idea of building bio-inspired cognitive adaptive solid-state devices has been around for decades. It forms the basis for synaptic electronics, a field of research that aims to build artificial synaptic devices to emulate the computation performed by biological synapses.

Synapses dominate the architecture of the brain and are responsible for massive parallelism, structural plasticity and robustness of the brain. These are also crucial for biological computations that are needed for perception and learning. Therefore a compact nano-electronic device emulating the functions and plasticity of biological synapses will be the most important building block of brain-inspired computational systems.

Scientific interest has expanded



Electrical pulses convert a doped chalcogenide from an amorphous phase (pink) into a crystalline one (blue). Once a certain amount has been converted, conductance of the material suddenly jumps, mimicking the firing of a neuron (Image courtesy: cen.acs.org)

towards building electronic systems that mimic the ability of the human brain in performing energy-efficient and fault-tolerant computation in a compact space, and research activities in this field have been growing rapidly.

Various efforts in neuromorphic engineering—a new interdisciplinary field that includes nanotechnologies—are being carried out, and the goal is to design artificial neural systems with physical architectures similar to biological nervous systems. In research findings, scientists have developed a carbon nanotube synapse with the elementary dynamic logic, learning and memory functions of a biological synapse.

Research efforts on brain-inspired computing and synaptic electronics can be understood considering the inefficiency of conventional computational systems in solving complex problems. Researchers are focusing on developing synapse-like electronic

devices with directed attention to material systems that have been investigated for non-volatile memory technologies.

Interconnection scheme of phase-change memory (PCM) synapses to reach ultra-high density and compactness of brain is being developed. In the crossbar array architecture, PCM synapses lie between post-spike and pre-spike electrodes, inspired by biological synapses formed between pre-synaptic and post-synaptic neurons. A broad spectrum of device systems with programmable conductance inspired by already existing device technologies, such as PCM, resistive-change memory, conductive bridge-type memory, ferroelectric switches, carbon nanotube devices and three terminal devices or field effect transistors based devices, have been explored.

Using pectin. Memristors can provide a bridge for interfacing electronic circuits with nervous systems, moving us closer to the